

LazyFCA Classification of Telco Customers Churn

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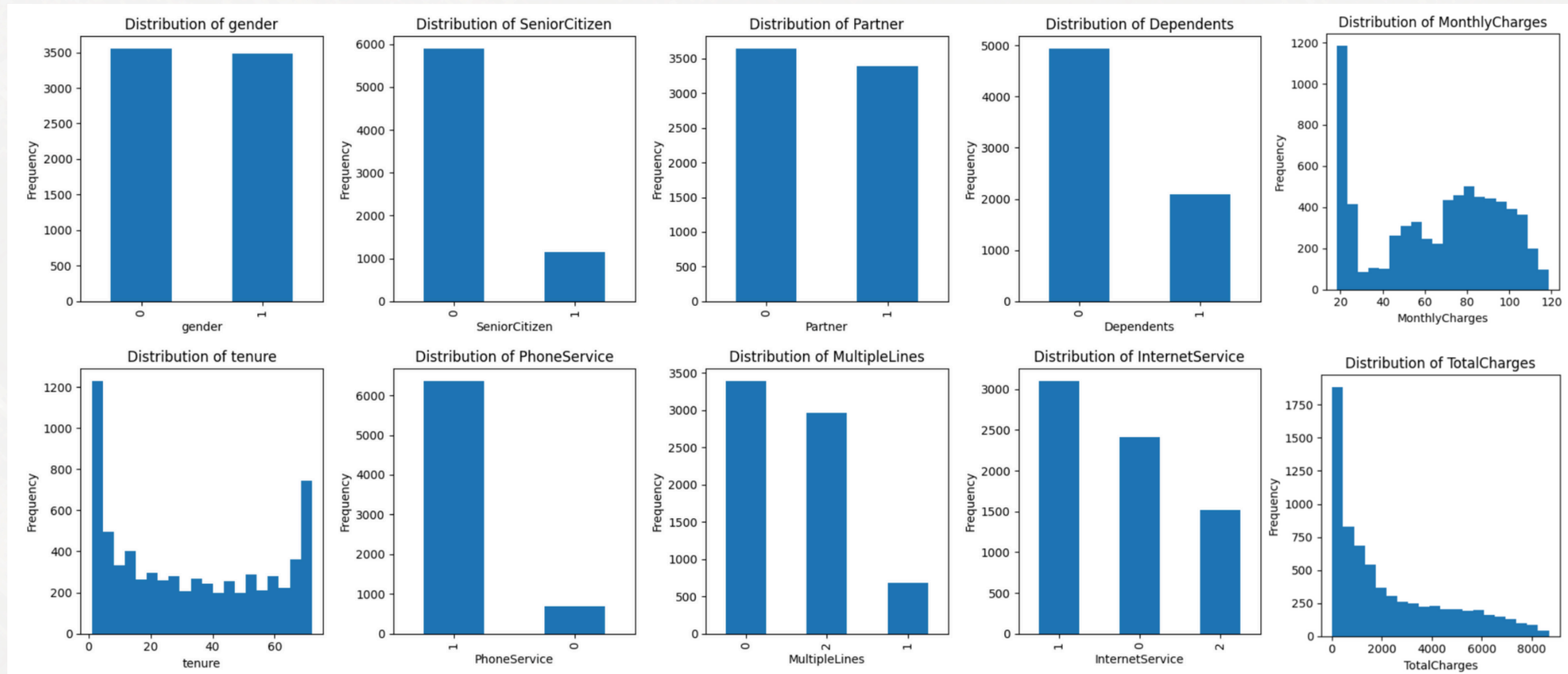
Dataset overview

- 1 Dataset from Kaggle: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>
- 2 7032 objects, 21 column
- 3 Example of data:

customerID	gender	Senior Citizen	Partner	Dependents	tenure	Phone Service	Multiple Lines	...	Churn
7590-VHVEG	Female	0	Yes	No	1	No	No phone service	...	No
55575-GNVDE	Male	0	No	No	34	Yes	No	...	No
3668-QPYBK	Male	0	No	No	2	Yes	No	...	Yes
7795-CFOCW	Male	0	No	No	45	No	No phone service	...	No
1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	...	No

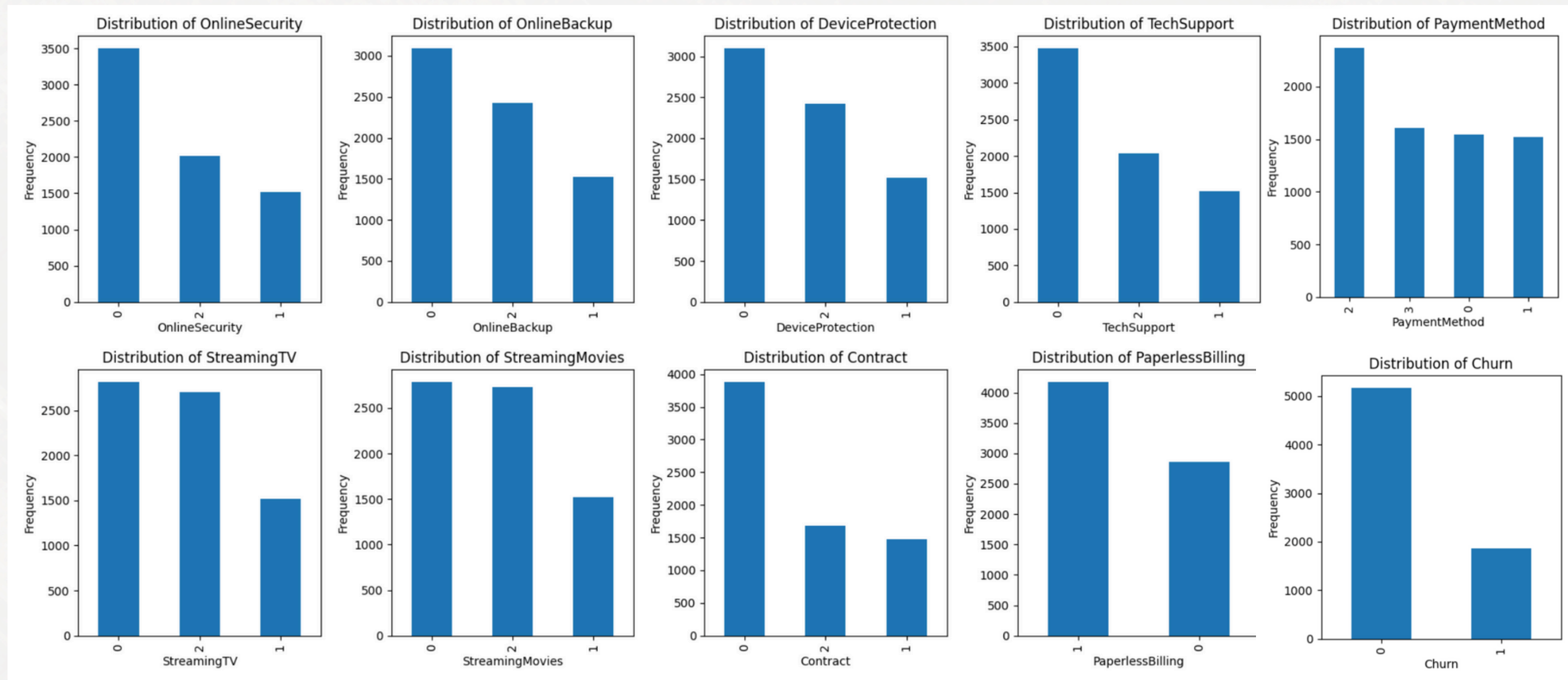
Dataset overview

Distribution of columns (before feature removal, under sampling, normalisation, one-hot encoding):



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Classic models results

Parameters were fine-tuned using GridSearchCV

	TP	TN	FP	FN	TNR	NPV	FPR	FDR	Accuracy	Precision	Recall	F1
KNN	275	269	105	99	0.72	0.73	0.28	0.28	0.73	0.72	0.74	0.73
NaiveBayes	326	212	162	48	0.57	0.81	0.43	0.33	0.72	0.67	0.87	0.76
LogReg	280	277	97	94	0.74	0.75	0.26	0.26	0.75	0.74	0.75	0.75
SVM	288	269	105	86	0.72	0.76	0.28	0.27	0.75	0.73	0.77	0.75
DecisionTree	326	216	158	48	0.58	0.82	0.42	0.33	0.73	0.67	0.87	0.76
RandomForest	292	268	106	82	0.72	0.77	0.28	0.27	0.75	0.73	0.78	0.76
XGBoost	290	270	104	84	0.72	0.76	0.28	0.26	0.75	0.74	0.78	0.76

Classic models vs LazyFCA results

	TP	TN	FP	FN	TNR	NPV	FPR	FDR	Accuracy	Precision	Recall	F1
KNN	275	269	105	99	0.72	0.73	0.28	0.28	0.73	0.72	0.74	0.73
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LazyFCA	277	266	108	97	0.71	0.73	0.29	0.28	0.73	0.72	0.74	0.73

— Lazy FCA improvements

- ① GPU acceleration
- ② Random sample: 50%
- ③ Redefining the rule of determining the class (with parameters tested through GridSearch):

```
if negative_classifiers / (negative_classifiers + positive_classifiers) < alpha and positive_classifiers >= beta:  
    return 1 # positive  
elif negative_classifiers / (negative_classifiers + positive_classifiers) > alpha and negative_classifiers >= beta:  
    return 0 # negative  
else:  
    return 1
```

— Lazy FCA improvements results

	TP	TN	FP	FN	TNR	NPV	FPR	FDR	Accuracy	Precision	Recall	F1
KNN	275	269	105	99	0.72	0.73	0.28	0.28	0.73	0.72	0.74	0.73
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LazyFCA	277	266	108	97	0.71	0.73	0.29	0.28	0.73	0.72	0.74	0.73
LazyFCA improved	321	212	162	53	0.57	0.8	0.43	0.34	0.71	0.67	0.86	0.75

Interpretability analysis

① Averaged explanations over the entire dataset

- Average proportion of positive matches: 52.9%
- Average proportion of negative matches: 47.1%
- Average confidence: 75.06%

② Explanations for specific sample

Explanations for sample 97:

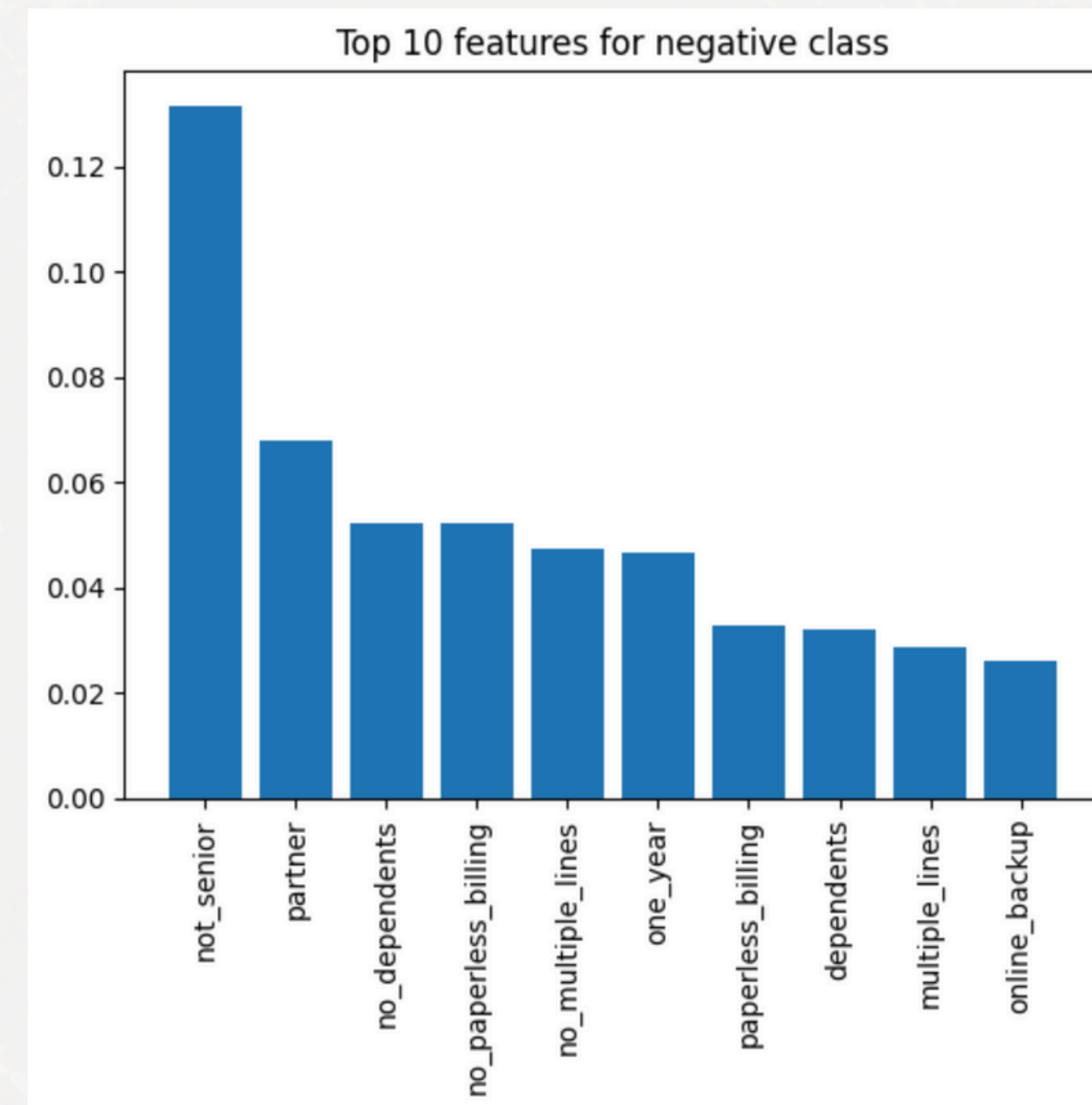
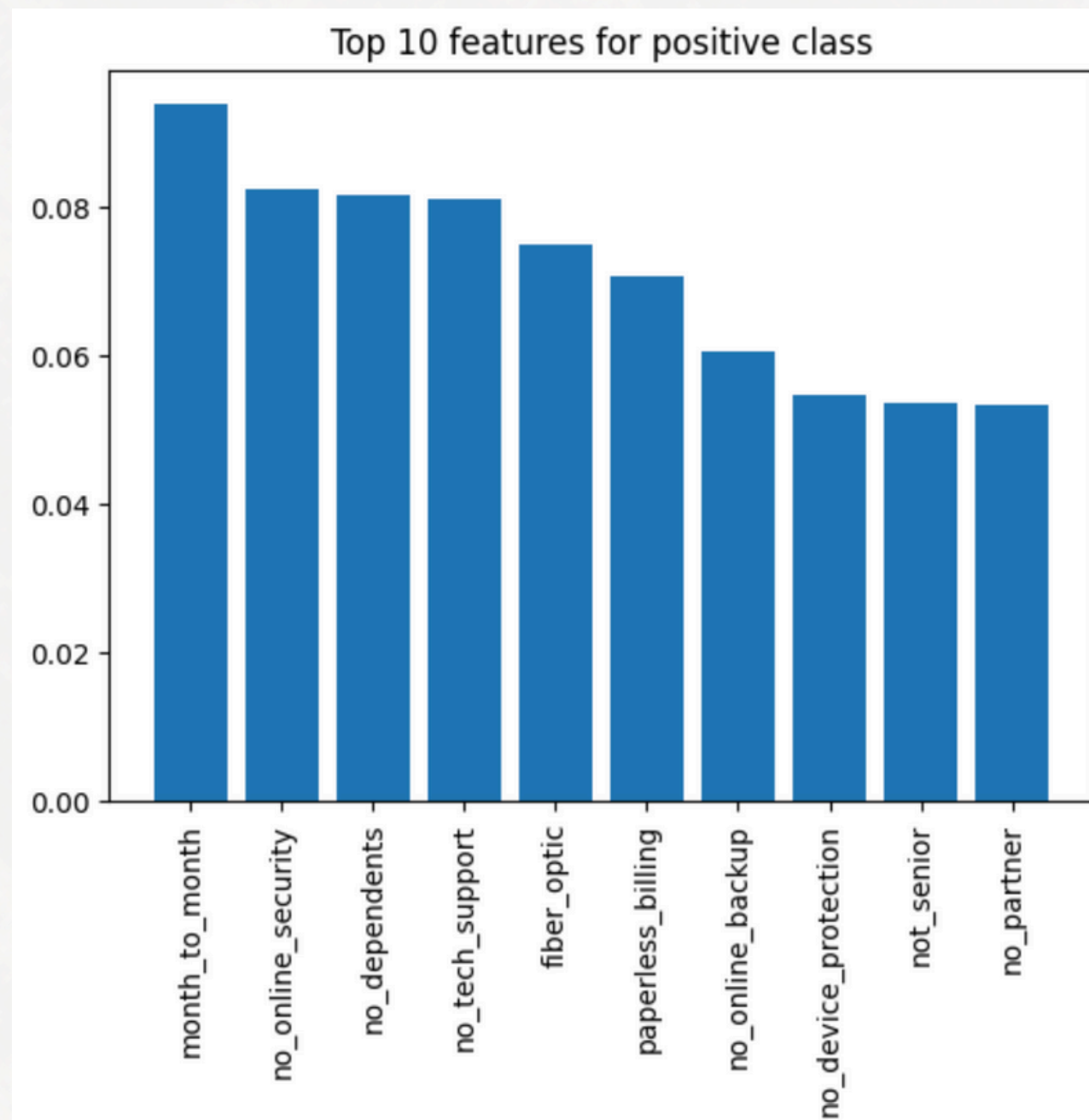
This sample is classified as POSITIVE with 75.3% confidence. It has 140 supporting classifiers and 46 opposing classifiers.

Key binary feature: 'fiber_optic' appears in 69% of supporters.
Strongest supporter has 18 matching samples.

To flip the prediction, the most impactful change would be: Flip no_online_security from True to False

Interpretability analysis

- ③ Most influential features for positive and negative classes



Conclusions

- ① Introduced modifications help to increase the speed of the classification procedure as well as to enhance the F1-score (through greater emphasis on Recall).
- ② The model tends to provide both global explanations as well as the local ones for specific samples that unravels the algorithms underlying rules for classification.

Future improvements

Future studies could focus on introducing new modifications that would allow to raise Accuracy and Precision metrics as well (making the architecture more robust to various datasets for classification problems).

If you have any questions, we are ready to answer them